

It's time to move on from LLMs

The sheer number of AI announcements that punctuate our daily lives is just ridiculous. I don't even know if I should liken AI announcements to punctuation. That would be akin to describing an overabundance of punctuation. Imagine someone so painfully anal about commas that they have one after each word in a sentence. It, can, get, really, tedious, after, a, while. A lot of the announcements aren't even about AI. Some of them are using AI just as a marketing buzzword to get some eyeballs. A large chunk of today's 'AI' announcements can be brushed off simply because they're utilising more of a statistical model rather than neural networks. And there's a difference.

Statistical models have been around for decades and they've been instrumental in analysing and interpreting data long before the advent of modern computing. Rooted in the principles of probability theory and statistics, these models aim to describe the relationships between variables and make inferences about the population of data using a sample set. Example, you know that two out of fifteen people around you have developed some form of heart disease and, therefore, heart disease might affect 2/15 i.e. 13.33% of the human population. That's inferring a conclusion about a population i.e. the human race, using a sample set i.e. the 15 people around you. Your accuracy might be ridiculously off and you've not considered a hundred other parameters such as age, physical fitness, vices, etc that impact your deduction, however, you've got a slight inkling about the health statistics of people in the world. This statistical model doesn't improve until you start incorporating more parameters and rewrite your methods for drawing inferences. That's how statistical models are.

Machine learning, on the other hand, does not need to be explicitly programmed. Machine learning is all about building systems that can learn from data and improve their performance over time and this domain encompasses a broader range of techniques and applications compared to traditional statistical models. Unlike statistical models, machine learning methods are often non-parametric which means that they do not assume a fixed number of parameters. In



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fact, they can adapt to the complexity of the data. This is why AI is often described as a black box.

So what are LLMs (Large Language Models)? Are they statistical models or machine learning? LLMs rely on breaking text down into smaller parts called tokens. Also, they use a machine learning model known as a Transformer. Transformer models are trained on vast amounts of text data to predict the next word in a sequence. The focus is more on the relationships between words in a sentence and sentences in a paragraph. There complex patterns is what help LLMs figure out which word follows another in a sentence. If you feed the transformer a book on grammar and 50 poorly written essays with similar glaring mistakes, then the transformer is going to be really bad at grammar. If you suddenly start getting poor responses from your LLM, you know that you've fed it bad data and that you need to go back and revert to an earlier state. Alas, most LLMs aren't fixed this way. They're fixed by throwing more data at it, thereby, making it statistically better at grammar. This literally means that the model will continue to make mistakes.

This is why LLMs are so bad at reasoning or why they hallucinate. It just looks at the relationships of the words rather than the proper context. Also, this is why you need to give LLMs a lot of prompts to get what you want. If you're satisfied with an LLMs initial response, then either you've got an accurate LLM or you can't tell the mistakes in the output because you don't know better.

Unfortunately, the hype around LLMs means that some of the cutting edge AI work gets no attention. Don't get me wrong, LLMs are a remarkable achievement in the field of machine learning but they represent just one aspect. Most of the coverage is about LLMs and I'm very cognizant of the fact that we too contribute to hero-ing LLMs and not highlight other noteworthy AI developments. We're actively working towards changing that. Let me start off by pointing out 3D Gaussian Splatting. It's an algorithm for Neural Radiance Field (NeRF) which allows quicker production of 3D videos using 2D photos. You'll see its applications in generative AI video creation tools within this year. And we'll make sure to give it its rightful share of the limelight. ■